

Shrub Hub

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Overview

This document contains the necessary information to build the Shrub Hub, a low cost, easy to build digital switch interface.



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Maker Checklist

This list provides an overview of the steps required to build and deliver the Shrub Hub.

Maker To Do List

- ☐ Read through the Maker Guide to become familiar with required components, tools, supplies, safety gear, and overall assembly steps.
- ☐ Talk to the User about how many switches they want to use
 - ☐ The Switch Adapter is only needed for 2 to 3 switches, if the User only wants 1 switch, you do not need to build the adapter
- ☐ Talk to the User about customization options
 - ☐ Colour of printed enclosure
 - ☐ How they would like to receive the “User Guide” (PDF or physical copy)
 - ☐ If they need the Switch Adapter, how long do they want the cables?
 - ☐ Do they need a USB extender?
 - ☐ Do they need the outputs of the device to be changed?
- ☐ Order hardware components
- ☐ Gather tools, supplies, and safety equipment.
- ☐ Assemble the device
- ☐ Flash firmware to Shrub Hub
- ☐ Test the Shrub Hub
- ☐ Print “User Guide” (if the User would like a physical copy)

Items to Give to User

- ☐ Shrub Hub
- ☐ Switch Adapter (if required)
- ☐ “User Guide”

Component Sourcing Notes

The 3-to-1 Switch Adapter is designed for mono audio cables with an outer diameter of 3 mm like the ones specified in the Bill of Materials. Cables with larger diameters may not work and may cause the enclosure to break.

The 3-to-1 Switch Adapter is designed for the TRRS terminal block sourced from Amazon. A similar part is available from Adafruit, but is too small and will not fit within the enclosure.

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Tool List

Tools / Equipment

Tool ID	Description	Required / Recommended	Notes
T01	Wire Strippers	Recommended	Stripping mono cable wires.
T02	Small Phillips Screwdriver	Required	Operating TRRS terminal screws
T03	Silver Sharpie® Marker	Recommended	Marking each cable

Customization Guide

The device can be printed in the user's desired colour.

The device outputs can be changed to better suit the user's needs

3D Printing Guide

The device was originally printed on a Bambu P1S using BambuStudio slicer.

3D Printing Summary

Metrics	Single Unit
Total Print Time (hour:min)	0:32
Total Number of Components	4
Typical Total Mass (g)	11
Typical Number of Print Setups	1

3D Printing Settings

Note that the 3D printing material should be assumed to be PLA unless otherwise noted in the table below.

Print File Name	Qty	Total Print Time (hr:min)	Mass (g)	Infill (%)	Support(Y/N)	Layer Height/ Nozzle Diameter(mm)	Notes
Shrub Hub Base.stl	1	0:06	3	20	N	0.2/0.4	
Shrub Hub Lid.stl	1	0:05	1	20	N	0.2/0.4	
Switch Adapter Base.stl	1	0:08	4	20	N	0.2/0.4	
Switch Adapter Top.stl	1	0:06	3	20	N	0.2/0.4	

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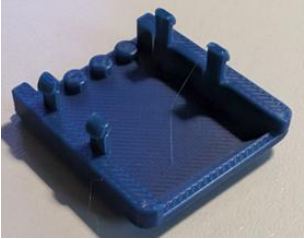
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Post-Processing

Inspect the 3D printed parts for any printing defects, sharp edges, or burrs. Sharp edges and burrs can be removed with sanding or deburring tools.

Examples of Quality Prints

Compare your 3D prints to the images here. If there are significant differences, you may need to reprint the part.

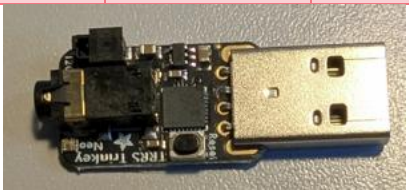
Shrub Hub 3D Prints		
Shrub Hub Base	Shrub Hub Lid	Switch Adapter Base
		
Switch Adapter Top		
		

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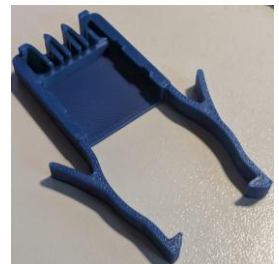
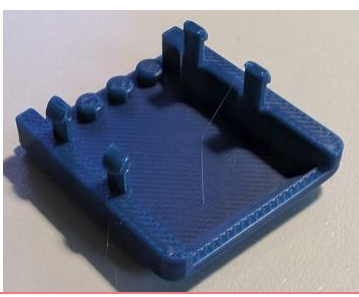
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Maker Component List

Part A: Shrub Hub Components

A01	TRRS Trinkey	QTY: 1	A02	Shrub Hub Base	QTY: 1	A03	Shrub Hub Lid	QTY: 1
								
A04	Light Pipe	QTY: 1						
								

Part B: 3-to-1 Switch Adapter Components

B01	TRRS Terminal	QTY: 1	B02	Female Mono Cable	QTY: 3	B03	Switch Adapter Base	QTY: 1
								
B04	Switch Adapter Top	QTY: 1						
								

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Assembly Guide

Part A: Assemble the Shrub Hub

Part A: Required Tools and Supplies

- None

Part A: Shrub Hub Assembly Steps



Step A-01: Insert the light pipe into the Shrub Hub Lid

Take the light pipe (A04) and press it into the hole on the Shrub Hub Lid (A03).

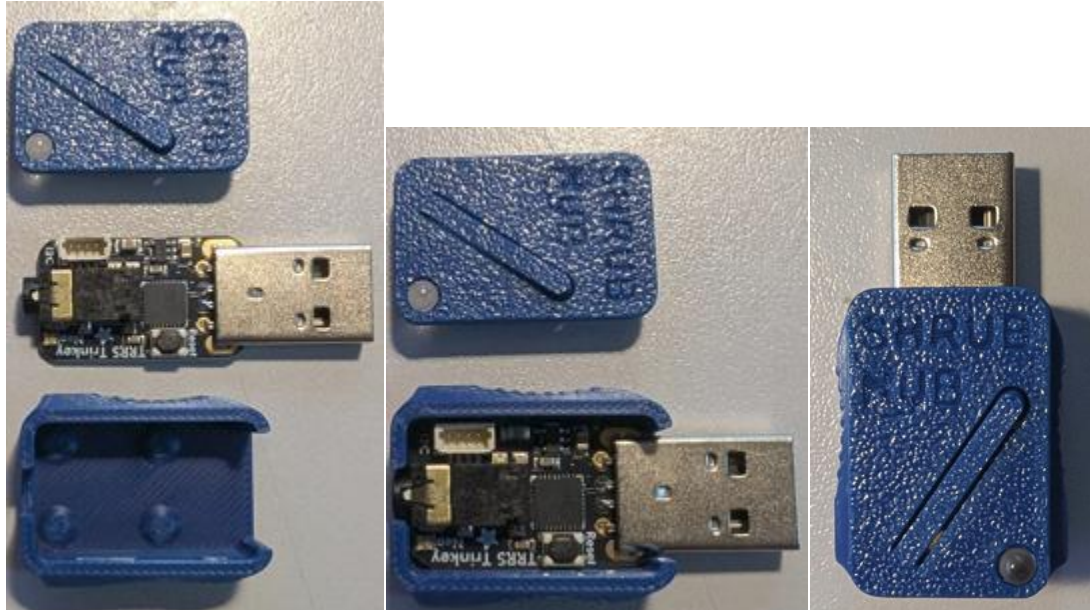


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Step A-02: Attach the enclosure to the Trinkey

Place the TRRS Trinkey (A01) into the Shrub Hub Base (A02), then snap the Shrub Hub Lid (A03) on top of it to lock the Trinkey in place.



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Part B: Assemble the Cable Adapter

Part B: Required Tools and Supplies

- Small Phillips Screwdriver
- Wire Strippers
- Silver Sharpie® Marker

Part B: Cable Adapter Assembly Steps



Step B-01: Strip the mono cables

Using wire strippers (T01), strip the ends on each mono cable (B02) so there is 1 cm of wire exposed. The wires do come pre stripped, but having slightly more wire available makes the next step easier. If the user wants a shorter cable, trim the entire cable shorter before exposing the wires and stripping the ends.



Step B-02: Connect the Mono Cables to the TRRS Terminal Jack

Using the Small Phillips Screwdriver (T02), open the terminals on the TRRS Terminal Jack (B01). Put the black wires from all three of the stripped mono cables into the terminal marked L and then tighten the terminal to lock all the wires in place. Take the red wire from each of the mono cables, and attach one to each of the remaining terminals, one to R, one to V, and one to the ground (arrow shaped mark).

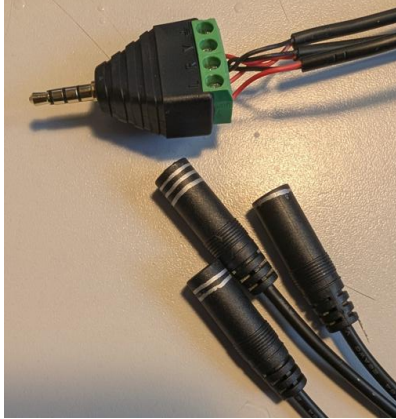


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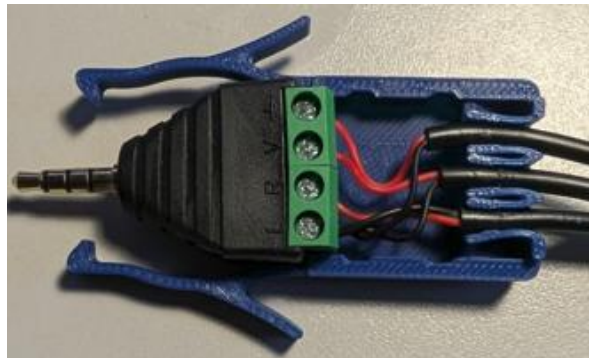
Step B-03: Mark the Mono Cable connectors

Using a silver Sharpie® marker (T03) or other marking device, label the cable plug for Switch 1, Switch 2, and Switch 3. Switch 1 is the one with the red wire connecting to the V terminal, Switch 2 connects to the R terminal, and Switch 3 connects to the ground terminal.



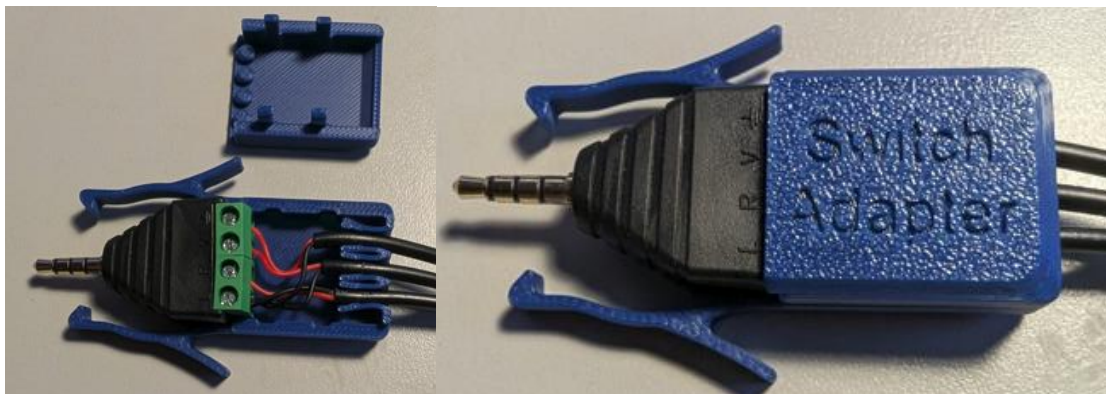
Step B-04: Assemble the Switch Adapter Base

Take the Switch Adapter Base (B03) and place the TRRS splitter jack between the gripping arms of the Switch Adapter Base. Press the mono cable wires one by one into the retention slots, pushing until they are at the bottom of the slot. Leave some slack in the wires that are screwed into the terminal block.



Step B-05: Close the Switch Adapter

Take the Switch Adapter Lid (B04) and press it into the base, aligning the four tabs with the slots on the base, and the four posts with the screw holes on the terminal block.



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Part C: Flashing Firmware to Shrub Hub

Part C: Required Tools and Supplies

- Computer that can open a USB flash drive

Part C: Programming Steps

Step C-01: Connect the Shrub Hub to the Computer

Plug the Shrub Hub into a USB port on the computer. If the computer does not have a USB-A port, an adapter will be required to connect the Hub to the computer.

Step C-02: Install CircuitPython

1. Open File Explorer
2. Double press the reset button on the Shrub Hub. A drive called TRINKEYBOOT should appear, similar to plugging in a USB flash drive.
3. Open the folder in the /Build_Files/Firmware_Files/Bootloader directory in the project files for the Shrub Hub.
4. Drag and drop the provided .uf2 file into the drive. It should then disappear, and a drive called CIRCUITPY should appear.

Step C-03: Add the library files

1. Open the lib folder in the /Build_Files/Firmware_Files/Shrub Hub Firmware directory
2. Copy all the files found there.
3. Paste them into the **lib folder** in the CIRCUITPY drive

Name	Date modified	Type	Size
adafruit_hid	2025-07-29 11:56 AM	File folder	
adafruit_pixelbuf.mpy	2025-07-29 2:25 PM	MPY File	3 KB
neopixel.mpy	2025-07-29 2:25 PM	MPY File	2 KB

Note: If you downloaded individual files instead of the entire project repository, it is recommended to download the repository for this step. Copying the contents of the folder onto the Shrub Hub is much easier than downloading each file individually and recreating the file structure yourself. The repository can be downloaded from the device listing on the MMC website or by clicking on the green <>Code button on the Shrub Hub GitHub page then the Download Zip button on the popup.

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Step C-04: Program the Shrub Hub

1. Copy the code.py file from the /Build_Files/Firmware_Files/Shrub Hub Firmware directory
2. Paste it into the CIRCUITPY drive, replacing the code.py file already there

Optional: Changing the Shrub Hub Outputs

The outputs of each mode can be changed to better suit the needs of the user. The instructions for how to do this can be found in the Changing Outputs Guide in the Shrub Hub documentation folder. This requires opening and modifying the code, and should only be done by someone who is confident in their ability.

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Testing Guide

Once the Shrub Hub is programmed, connect assistive switches to each of the 3.5 mm inputs of the switch adapter, and connect the switch adapter to the Shrub Hub. Test that each switch produces the correct output, and that holding any switch for more than 4 seconds changes the mode. Mark each switch port with the correct switch input number if not already done.



Input		Output				
Activation	Input Switch	Mode 1 (Teal)	Mode 2 (Purple)	Mode 3 (Yellow)	Mode 4 (Green)	Mode 5 (Orange)
One Switch	Switch 1	Enter	F1	Left Click	F13	Play/Pause
	Switch 2	Space	F2	Right Click	F14	Volume Up
	Switch 3	Tab	F3	Middle Click	F15	Volume Down
Two Switch	Switch 1+2	Backspace	F4	Left Click	F16	Next Track
	Switch 2+3	Escape	F5	Right Click	F17	Previous Track
	Switch 1+3	Enter	F1	Left Click	F13	Play/Pause
Three Switch	Switch 1+2+3	Enter	F1	Left Click	F13	Play/Pause

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Troubleshooting Guide

	Problem, Issue, or Concern	Cause	Remedy
1.1	Nothing happens when switch is activated	Switch isn't properly connected	Confirm that each switch is firmly connected to the intended input port.
1.2		Switch isn't working	Confirm that the switch is working with a different device or switch tester.
1.3		A non-noticeable output is generated from a different operating mode than expected	Check the Mode Indicator to ensure that the Shrub Hub is in the desired operating mode. If not, Change Operating Mode.
1.4		Loose or disconnected Switch Adapter	Confirm that the Switch Adapter is securely connected to the Shrub Hub.
2.1	Unexpected output is generated when switch is activated	Different switch activated	Confirm each switch is connected to the intended input port
2.2		Different operating mode than expected	Check the Mode Indicator to ensure that the Shrub Hub is in the desired operating mode. If not, Change Operating Mode.
3.1	Mode Indicator is blinking yellow	Host device does not support USB HID	Connect Shrub Hub to a device that supports USB HID
3.2		Shrub Hub connected to USB Power Adapter	Connect Shrub Hub to a device that supports USB HID
4.1	Mode Indicator is blinking red	The Shrub Hub code is not compiling	Revert the code to the latest version found on the Shrub Hub GitHub.
5.1	The Shrub Hub freezes and does not respond to button presses	Shorted cables in Switch Adapter	Open the Switch Adapter enclosure and ensure there is no short between two of the cables.

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Support and Resources

Getting Help

The device referred to in this guide is an open-source assistive technology and is supported by a combination of documentation and community support.

If you need help building this device, please start by reviewing the documentation, including the



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Troubleshooting Guide. If you are still having trouble, you can reach out for support in several ways:

1. Email the Makers Making Change Team at info@makersmakingchange.com.
2. Reach out to Makers Making Change on social media.

When asking for help, it is helpful to include:

- The name of the device
- A brief description of the issue
- What you were trying to do when the problem occurred
- Photos or videos, if available

Project Page & Documentation

The project page contains the most up-to-date information about this device, including documentation, design files, and known issues.

- Project page: <https://github.com/makersmakingchange/Shrub-Hub>

Reporting Issues or Suggesting Improvements

If you want to report a problem with the Shrub Hub, or suggest an improvement, you can get in touch in the following ways:

- Leave an issue on the GitHub repository, found at
 - <https://github.com/makersmakingchange/Shrub-Hub/issues>
- Send us an email at info@makersmakingchange.com

Open Source & Licensing

The device referred to in this guide is open-source assistive technology. This means the design files and documentation are publicly available and may be used, modified, and shared under the terms of the project's license.

Open-source devices are often built and adapted by volunteers. While care is taken to provide accurate documentation, performance cannot be guaranteed in every situation. Always use your judgment and consult a professional if you have questions about suitability or safety.